

AI-Assisted Data Analysis Report (Python + SQL)

1. Introduction (Theory)

Data analysis is the process of extracting meaningful insights from raw data. In modern analytics, tools like Python and SQL are widely used for data cleaning, transformation, and analysis. Artificial Intelligence (AI) tools such as ChatGPT help in improving code quality, reducing development time, and enhancing analytical capabilities.

2. Python & Pandas (Theory)

Pandas is a powerful Python library used for data manipulation and analysis. It allows users to handle structured data efficiently using DataFrames. Key operations include handling missing values, filtering data, grouping, and aggregation.

3. Feature Engineering (Theory)

Feature engineering is the process of creating new variables from existing data to improve analysis. In this project, features such as total_spent, total_orders, average_order_value, and days_since_last_purchase were created to better understand customer behavior.

4. SQL Analysis (Theory)

SQL (Structured Query Language) is used to query and analyze relational databases. Advanced SQL techniques such as JOINS, GROUP BY, and window functions help in extracting meaningful insights from large datasets.

5. Executive Summary

This report presents an AI-assisted data analysis using Python and SQL. Data was cleaned, transformed, and analyzed to generate meaningful insights. Customer behavior was analyzed using features such as total spending and purchase frequency. SQL queries were used to extract business insights such as revenue trends and customer segmentation. The integration of AI tools improved efficiency and accuracy in the analysis process.

6. Python Code (Sample)

```
customer_features = transactions.groupby("customer_id").agg(
    total_spent=("total_amount", "sum"),
    total_orders=("order_id", "count"),
    avg_order_value=("total_amount", "mean")
).reset_index()
```

7. Sample Output

customer_segment	total_customers	total_revenue
Regular	120	250000
Occasional	80	90000
VIP	30	300000